RESEARCH REPORT · VITKI

Fully tested finding

Women raised runestones more often in Svealand than in Götaland – 31 against 17 percent

Viktor Kvant

Independent researcher · Vitki

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Timeline of the study: The pattern was observed and published 13 October 2026 · The analysis plan was locked 28 October 2026 · The follow-up plan was locked 29 October 2026 · The material was frozen 29 October 2026

ABSTRACT

Background and aim. That women raised runestones is known, and that their share varies between provinces has been observed before, also in two of my articles before the study. The question here is whether the difference between Svealand and Götaland remains when the age of the stones and the habits of individual carvers are held constant, and whether it withstands a fixed series of sensitivity analyses.

Material. 1,425 legible Swedish Viking-Age runestones with at least one named or mentioned sponsor, from the Scandinavian Runic-text Database (Rundata): 1,013 in Svealand and 370 in Götaland. The sponsors' sex is determined by rules checked against hand coding.

Method. Pre-registered re-test with logistic regression, region and dating period as explanatory variables and the carver as a random effect; Holm correction over the study's six tests; internal replication in two pre-defined halves of the material; seven sensitivity analyses; adjustment for all 48 associations the project reports (Benjamini–Hochberg).

Results. A woman among the sponsors on 31 percent of the stones in Svealand against 17 percent in Götaland; odds ratio Götaland against Svealand 0.46 (95 % interval 0.32–0.66), $p < 0.001$, Holm-corrected $p < 0.001$. Same direction and significant in both halves, and every sensitivity analysis holds; adjusted for all associations $q < 0.001$. A follow-up pre-registered study could not tie the difference to any role. It is clearest on stones for a child, where the mother appears on 55 against 19 percent of the stones raw (51 against 19 with time and carver constant), but that test holds in one half only and is an indication. For widows, lone women and Christian features no difference between the regions was shown.

Conclusion. The difference is large and withstands every check, but it is carried to a large degree by Uppland, and it measures who is named among the sponsors – a custom of the memorial stones – not women's standing. What it consists in is not established.

Keywords: *runestones, Viking Age, women, sponsors, Svealand, Götaland, logistic regression, pre-registration*

HIGHLIGHTS

- In Svealand a woman was named among those who raised the stone nearly twice as often as in Götaland, and the difference withstands every check in the study.
- No difference was shown for widows or lone women; the difference is clearest where the mother appears on the stone for her child – but that is an indication, not a finding.
- The difference concerns who is named on the memorial stones, not women's power – and it is carried to a large degree by Uppland.

1. Introduction

The women of the runestones have been a field of their own since the 1990s. Judith Jesch (1991) gathered what the inscriptions say about women, and Birgit Sawyer (2000) counted the sponsors province by province and showed that the share of women varies between provinces. Sawyer read the stones as documents of inheritance: whoever raised the stone presented herself or himself as the heir, and where women could inherit they appeared more often. Anne-Sofie Gräslund (1995) has discussed the same women in relation to Christianisation. The counts in that literature are raw shares. They take no account of the fact that the stones of Svealand are on average younger than those of Götaland, or that a few carvers account for many stones in Uppland.

The aim is to test whether the difference between Svealand and Götaland holds when the age of the stones and the carvers are held constant, with a model fixed before it was run, and to let it pass the same sensitivity analyses as every other association in the project. The study says nothing new about why women raised stones. It tests whether the regional difference is a finding to build on.

2. Pre-registration: hypothesis and plan

The hypothesis, the model and the rules of interpretation were written down in an analysis plan on 28 October 2026, before any model had been run. The plan concerned a larger study of who raised the runestones and for whom, with six tests; this is the first. The plan is time-stamped in the project's version control and has the SHA-256 checksum `ea26a388379e29eb…`, so that any later change would show (see Data and code).

The test reads in the plan: the outcome is at least one woman among the sponsors, among stones with at least one identified sponsor; the model is outcome ~ region + dating period + carver as a random effect; the measure is the odds ratio Götaland against Svealand. The six tests are corrected together by Holm's method. The rules of interpretation were fixed in advance: a test *confirms* when Holm p is below 0.05 and the direction is the same with p below 0.05 in each of the material's two

halves; a *pattern to test* has Holm p below 0.05 but does not hold in both halves; an *indication* has only uncorrected p below 0.05.

Before the plan was locked I had seen and published a raw comparison between the provinces: in the article She inherited from her child (13 October 2026) a woman was among the sponsors on 31 percent of the stones in Uppland against 11–20 percent in Östergötland, Västergötland and Småland, and The daughters the runestones forgot (14 October 2026) said that in Svealand the mother is on half of the stones for a child. On the women, the plan mentions only the figure 391. The main test and the role tests are thus pre-registered re-tests of patterns already seen, on the same material.

The analyses in sections 5.1–5.3 are those the plan prescribes. The follow-up in 5.4 comes from a second analysis plan, written and locked on 29 October 2026 (SHA-256 `6a03f6fd1a5f50f5…`) after the main result was known. It is thus pre-registered but not part of the first test, and it is reported as a follow-up. No exploratory analyses are included.

3. Material

- **Source.** The Scandinavian Runic-text Database (Rundata), Department of Scandinavian Languages, Uppsala University: transliteration, normalisation, translation, province, parish, carver and style group for each inscription. The material was frozen for the analysis on 29 October 2026.
- **The stones.** All Viking-Age runestones in the Swedish provinces: 2,320 stones. Runestones only; other objects with runes are not included. Skåne, Blekinge and Halland, Danish in the Viking Age, are counted as Denmark in Rundata and are not included.
- **Selection.** Stones whose text is preserved well enough for the sponsors to be read, and where at least one sponsor can be identified: 1,425 stones. Of these, 1,013 lie in Svealand and 370 in Götaland. 42 lie on Gotland or in Norrland; they enter the model as their own levels but are not compared.
- **The regions** by today's provinces: Svealand is Uppland, Södermanland, Västmanland, Närke, Gästrikland and Värmland (Dalarna has no stones in the selection); Götaland is Östergötland, Västergötland, Småland, Öland and Bohuslän (Dalsland has no stones in the selection).
- **The outcome** is that at least one woman stands among the sponsors. The sponsors' sex is read from Rundata's English translation by rules in four steps: first the relation (mother, daughter, wife; son, father, brother), then a lone sponsor's possessive in the formula (*her son*, *his son*), then a name lexicon built from the first two steps, and last the name's ending. The rules were checked against hand coding on a new sample of 40 stones the rules had never seen: 39 of 40 right. In the hand-coded samples together (80 stones that could be compared) the rule's yes was right 13 times of 13 and its no 67 times of 67. Of the 80, 45 come from the first sample, which the rules were adjusted against; the one miscoding in the new sample (U 1090) is not among the comparable stones. The errors sit where the text has gaps, and they miss women more often than they add them.
- **Dating period** by the stone's ornamental style (Gräslund 2006): before 1010, 1010–1040, 1040–1070, after 1070, and undated as a group of its own.

- **The carver.** The carver Rundata gives. Stones without a known carver are grouped by parish.

4. Method

The statistical terms are explained in the reports' glossary: https://vitkirunor.se/en/forskningsrapporter/begrepp.

4.1 The model. Logistic regression with the R package mgcv (R 4.6.1; Wood 2017): the outcome against region and dating period, with the carver as a random effect. The random effect gives each carver a level of their own, so that a prolific carver's habits are not counted as a region's custom. The shares per region are the model's marginal effects, that is, the average over the stones as they are.

4.2 Correction and replication. Holm correction (Holm 1979) over the study's six tests. Internal replication: the test must hold in each of the material's two halves. The halves are formed with the same rule, fixed before the study, in all the project's analyses (whole parishes dealt alternately within each province in order of the parish's mean dating); since the rule is applied to the selection being tested, the size of the halves differs slightly between the tests. In addition, five robustness checks prescribed by the plan: only stones with at least ten legible words, only stones where every sponsor's sex is determined, the model without the carver effect, Uppland only, and sons against fathers (the last applies only to the study's tests about the dead).

4.3 Sensitivity analyses. The project applies the same fixed series of sensitivity analyses to every association it reports, whatever study the association comes from: the raw model; the model adjusted for dating period; the carver as a random effect; the parish as a random effect, to see whether the difference is the peculiarity of a few parishes; each half on its own; and the model refitted without one province at a time (provinces with at least 30 stones). An analysis with the stone's location as a spatial surface is not made here, since the comparison itself is geographic. An analysis holds when p is below 0.05 and the direction is the same as in the adjusted model. When a province is removed, significance is required only where at least two thirds of the stones remain; otherwise the direction suffices.

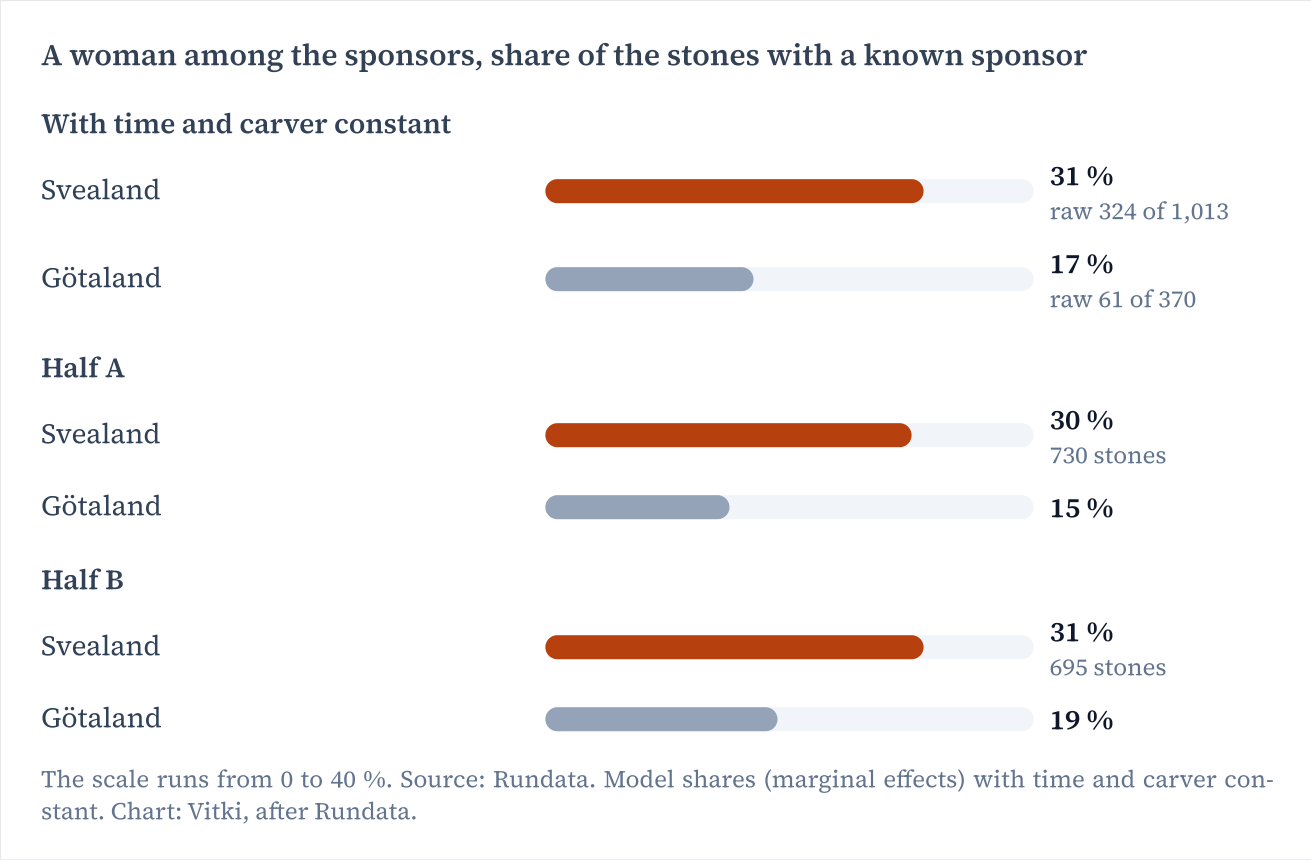
Since the project tests many associations, the p-values are also adjusted together for all 48 associations reported (Benjamini & Hochberg 1995). The adjusted value is called q below. An association counts as confirmed when it holds in both halves and q is below 0.05. The model's assumptions are checked with simulated residuals (DHARMa; Hartig 2022): the model simulates new outcomes for each stone many times, and each real outcome is compared with the simulated ones. The checks look for an uneven spread, more variation than the model allows for (overdispersion) and neighbouring stones that are more alike than the model explains (remaining spatial pattern). The errors in the sex determination are tested by refitting the model 200 times, with each stone's ”true” outcome drawn from the hit rates in the hand-coded sample.

The sensitivity analyses are computed on the 1,383 stones in Svealand and Götaland. The main model has 1,425, since Gotland and Norrland are included there as levels of their own. The sensitivity analyses use the project's common dating periods, so their figures can differ slightly from the main test's.

4.4 Deviations from the plan. None for the main plan. In the follow-up plan, one of the tests (a lone woman raises the stone) was first run with a variable that was missing from the data file, which gave zero stones. The variable was then computed exactly as the plan defines it and the test rerun. Nothing else was changed.

5. Results

5.1 The main test. Raw, a woman stands among the sponsors on 324 of 1,013 stones in Svealand (32 percent) against 61 of 370 in Götaland (16 percent). With time and carver constant the shares are 31 against 17 percent. The odds ratio Götaland against Svealand is 0.46 (95 % interval 0.32–0.66), $p < 0.001$, Holm-corrected $p < 0.001$. In plain words: draw a stone at random in each region, and it is nearly twice as common that the Svealand stone has a woman among the sponsors; a swing that large from chance alone is too rare to reckon with. Half A: 730 stones, odds ratio 0.41 (0.24–0.69), $p < 0.001$. Half B: 695 stones, odds ratio 0.51 (0.31–0.85), $p = 0.009$. Same direction and significant in both halves: the test *confirms* by the plan's rule.



5.2 The plan's robustness checks.

- Only stones with at least ten legible words: 951 stones, odds ratio 0.51, $p = 0.005$.
- Only stones where every sponsor's sex is determined: 1,390 stones, odds ratio 0.46, $p < 0.001$.
- Without the carver effect: odds ratio 0.46, $p < 0.001$.

- Uppland only: cannot be tested, since the comparison concerns regions and Uppland lies wholly in Svealand.

5.3 The sensitivity analyses (1,383 stones, 385 with a woman among the sponsors). All 7 hold:

- Raw model: odds ratio 0.42 (0.31–0.57), p < 0.001.
- Adjusted for dating period: 0.46 (0.32–0.65), p < 0.001.
- Carver as a random effect (483 carvers and parish groups): 0.45 (0.31–0.64), p < 0.001.
- Parish as a random effect (469 parishes): 0.46 (0.32–0.66), p < 0.001. The difference is not the peculiarity of a few parishes.
- Half A (701 stones): 0.43, p < 0.001. Half B (682 stones): 0.49, p = 0.004.
- Without one province at a time (6 provinces tested): the direction holds without each of them, and no province turns it. Weakest without Uppland, when 643 stones remain: odds ratio 0.66 (0.43–1.03), p = 0.07. Without Uppland more than half of Svealand's stones are gone, and there only the direction is required.
- The difference in percentage points in the adjusted model: 31 against 17 percent, 14 points (probably between 9 and 20).
- The model's assumptions: residual distribution p = 0.59, overdispersion p = 0.71, remaining spatial pattern p = 0.67. Nothing amiss.
- Errors in the sex determination: the association holds in 100 percent of the 200 refits; median odds ratio 0.47, range 0.44–0.55.
- Adjusted for all 48 associations the project reports: q < 0.001. The association is one of 15 that hold in every sensitivity analysis.

5.4 Follow-up (second pre-registered plan). Five pre-set explanations of what the difference consists of, tested with the same model and by the follow-up plan's rules of interpretation: a test *holds* when Holm p is below 0.05 and p is below 0.05 in the same direction in both halves; an *indication* has Holm p below 0.05 but does not hold in both halves, or has only uncorrected p below 0.05; otherwise no difference is shown. The woman's role is read from the relation to the dead: on a stone for a child she is the mother, on a stone for a father a daughter or wife, on a stone for a brother a sister or mother.

- **Widows after the journeys – not supported by the plan's rule.** A woman raises the stone for her husband: 96 of 1,013 in Svealand against 23 of 370 in Götaland, odds ratio 0.82 (95 % interval 0.47–1.43): no difference shown, but the interval does not rule out a moderate difference. Without the stones that mention journeys the difference between the regions remains and grows slightly (odds ratio 0.40, holds in both halves).
- **The custom of listing the family – not supported by the plan's rule; explains a fifth.** With the number of sponsors constant: odds ratio 0.53. 20 percent of the difference disappears, measured on the log-odds scale, that is, in the logarithm of the odds ratio, where the model is linear: the logarithm goes from −0.78 to −0.63. The plan's limit was half.
- **A position of her own – not supported by the plan's rule.** A lone woman raises the stone: 137 of 1,013 against 44 of 370, odds ratio 0.77 (95 % interval 0.50–1.18): no difference shown, but the interval also allows a moderate difference.

- **The roles – an indication, not established.** On stones for children (where the sex of the dead can be read nearly always sons, 220 of 235) the mother appears on 107 of 195 in Svealand (55 percent) against 13 of 70 in Götaland (19 percent), odds ratio 0.23, Holm-corrected p = 0.015, but in the halves p = 0.058 and 0.031: an indication by the follow-up plan's rule. On stones for fathers a daughter or wife appears on 66 of 415 against 9 of 117, odds ratio 0.37, Holm-corrected p = 0.12: an indication (uncorrected p = 0.02). On stones for brothers a sister or mother appears on 42 of 179 against 10 of 88 (22 against 13 percent with time and carver constant), odds ratio 0.53, p = 0.16: no difference shown. No role test holds, so by the plan the difference cannot be tied to a role.
- **Christianisation – no association.** A Christian feature on the stone (cross or prayer) and a woman among the sponsors: odds ratio 1.27, p = 0.16, and no difference between the regions.

6. Discussion

What the finding says. The difference between Svealand and Götaland is not due to Svealand's stones being younger or to a few carvers there carving many stones. It remains when both are held constant, in each half and in every sensitivity analysis. It is a difference in custom, not in population: a woman is more often *named* among those who raised the stone in Svealand. It does not say that women there had more power or property.

Against earlier research. Sawyer's (2000) observation that the share of women varies between provinces thus stands also when age and carver are held constant. The follow-up does not settle her interpretation. No difference between the regions in how often widows or lone women raised the stone was shown, but they are not ruled out (see the intervals in 5.4). The difference is clearest where the mother appears on the stone for her child, but that is an indication by the follow-up plan's rule. Sawyer read the stones as statements of inheritance; whether the mother pattern reflects inheritance, the journeys or something else is not tested here. The report rests on the main finding.

Weaknesses, in order of weight.

1. **Uppland carries much.** Without Uppland the odds ratio is 0.66 and the interval reaches above 1. The direction holds when any one province is removed, but the certain difference is to a large degree a difference between Uppland and Götaland.
2. **Sex is determined by rules.** The rules err on about one stone in twenty, mostly where the text has gaps, and they miss women more often than they add them. If gaps were more common in Götaland that would strengthen the difference on false grounds. With only stones of at least ten legible words the odds ratio is 0.51, so that does not explain the finding.
3. **The dating rests on ornamental style.** Undated stones are a group of their own. Another dating could move the figures slightly but hardly the direction, since the raw and the adjusted models nearly coincide.
4. **Region is a coarse division.** That the difference is largest between Uppland and Götaland is already visible. What happens in the border provinces is not tested here.

What would overturn the finding. A hand coding of sex on a large sample showing more errors in one region than in the other, or a recomputation with another dating in which the adjusted model no longer holds.

7. Conclusion

A woman stands among the sponsors on 31 percent of Svealand's runestones with a known sponsor against 17 percent of Götaland's, with age and carver constant, and the difference withstands every sensitivity analysis. What it consists in is not established: it is clearest where the mother appears on the stone for her child, but that is an indication, and no role test holds. The finding concerns who is named among the sponsors, a custom of the memorial stones, and is not a measure of women's standing. The roles can only be tested further on new material, and the question why the custom differed needs other material than the inscriptions' own words.

Data and code

The analysis was made in Vitki 3.0.0, an open research tool for runestones (vitkirunor.se). The analysis code as it stood when the results were computed is commit `5542b59` (29 October 2026) in https://github.com/Precatio/runes, the last change to the analysis and its results before report version 1; it is Vitki 3.0.0 with later changes. All code, the analysis plans and the result files are openly available there under GPL-3.0.

- The analysis plans: `analysplaner/2026-10-28-manniskorna.md` and `analysplaner/2026-10-29-kvinnorna-i-svealand.md`, with checksums in the `.sha256` files beside them.
- The sex determination: `src/people.py`; the hand coding: `data/people_check.json`.
- The main model: `r/people.R`. The sensitivity analyses: `r/robustness.R` and `src/robustness.py` (method description in `METHODS.md`, sections 1u and 1v). The results are saved under the material's version `e83202e01a5e6d27`.
- The follow-up: `utdata/kvinnor_landsdel/analys.R`, results and deviations in `utdata/kvinnor_landsdel/resultat.md`. It was run on the material's version `3f58e9339191761e`, which has the same stones and the same coding (identical `corpus.csv`, `people.csv` and `messages.csv`).
- To rerun the analysis: check out commit `5542b59`, start the analysis engine with `scripts/dev_api.sh` and run `curl -X POST http://localhost:8000/api/r/corpus/run`. That computes all the corpus analyses and the sensitivity analyses for the Rundata loaded and saves them under `data/cache/r/corpus/<version>/` (each R module is run as `Rscript --vanilla r/<module>.R params.json outdir`, see `METHODS.md`, section 1). The follow-up: `Rscript utdata/kvinnor_landsdel/analys.R data/cache/r/corpus/3f58e9339191761e outdir`.
- The report is built by `utdata/blogg/build/build_rapport_kvinnor.py` (`.venv/bin/python utdata/blogg/build/build_rapport_kvinnor.py`), which reads every figure from the result files and stops if anything disagrees; the builder as reviewed before archiving is in commit `6eae0af`.
- In Vitki the association is shown under Statistics, The people, with the sensitivity analyses step by step. The report applies to the material's version `e83202e01a5e6d27`. The app's latest version

(`3691a2c441dd03f4`) holds the Viking Age division constant in the sensitivity analyses (folklands, small lands and thirds, else the province) and so gives slightly different figures: weakest without one area at a time, the odds ratio is 0.54 (without Attundaland) instead of 0.66 (without Uppland). Every step holds there too. Popular accounts of the women on the stones are The daughters the runestones forgot and She inherited from her child.

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Change log

- **Version 1, 2026-10-31.** First published version.
- **Version 2, 2026-11-05.** Corrections after a review before archiving in Zenodo. The main test, its figures and the title are unchanged.
- *The pre-registration (section 2, the abstract, the timeline).* Version 1 said that no comparison between regions had been made before the plan was locked. That was wrong: two of my articles, of 13 and 14 October 2026, had already shown the pattern. The study is now described as a pre-registered re-test of a previously observed pattern, and the timeline begins 2026-10-13.
- *The roles (5.4, 6, 7, the abstract, the essentials box).* Version 1 said that the difference lies in the roles and that the mother appeared on her son's stone together with the father and the siblings. No role test holds by the follow-up plan's rule: the stones for children and for fathers are indications, and for the stones for brothers, which version 1 did not report, no difference was shown. That the mother appeared together with the father and the siblings was never tested. For widows and lone women the text now says that no difference was shown, with the intervals, instead of that they are ruled out.

- *Götaland's provinces (3).* Version 1 listed Halland and Dalsland. The selection has no stones from them; Halland's runestones are counted as Denmark in Rundata.
- *The number of sensitivity analyses.* The abstract said eight; they are seven.
- *Minor changes.* The weakness about Uppland now says ”when any one province is removed”; the origin of the 80 hand-coded stones is stated; simulated residuals and the log-odds scale are explained; the rule for the halves; the chart's source line; Rundata's version and licence; absolute links; the commit of the analysis and the commands to rerun it; a link to the glossary; a consistent voice.
- **Version 3, 2026-11-05.** Corrections after a second, independent reading of the changes in version 2. The main test, its figures and the title are unchanged.
- *The essentials box.* Version 2 said that widows and lone women ”do not explain it”. It now says, as the text does, that no difference was shown for them; the intervals do not rule them out.
- *Against earlier research (6).* Version 2 said that the mother pattern fits a rule of inheritance ”better than it fits the journeys or Christianisation”. No test in the plans compares those explanations; the text now says that whether the pattern reflects inheritance, the journeys or something else is not tested.
- *The custom of listing the family (5.4).* The heading now gives the plan's verdict, not supported, as for the other explanations.
- *Data and code.* The builder as reviewed before archiving is named with commit `6eae0af`.
- *What version 2 also changed without logging it.* Dalarna was removed from the list of Svealand's provinces (the selection has no stones from it). Data and code gained the follow-up's material version and the app's latest figures (weakest 0.54 without Attundaland), and the sentence on a future DOI was removed. In the English text ”raisers” became ”sponsors” in the lede and the chart's heading.

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Cite as

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https://vitkirunor.se/en/forskningsrapporter/kvinnor-bland-resarna-svealand-gotaland